

Clinical Features of PMLE

In any given patient, polymorphic light eruption (PMLE) outbreaks tend to affect the same exposed skin sites consistently, usually in a symmetric pattern. The extent of involvement may gradually spread or recede over time. Not all sun-exposed skin is affected—typically, only certain areas develop lesions, often those that are intermittently exposed or normally covered. Large regions of exposed skin may remain spared.

An extreme example is **juvenile spring eruption**, which primarily affects boys in the spring and presents with pruritic papules and vesicles localized to the ear helices. In some cases, typical PMLE may coexist.

Systemic symptoms are rare in PMLE, but isolated reports include malaise, nausea, and other nonspecific sensations.

Cutaneous Lesions

PMLE exhibits a wide range of morphologic presentations, all likely sharing similar pathogenesis and prognosis. Described forms include:

- **Papular** (most common)
- **Papulovesicular**
- **Plaque**
- **Vesiculobullous**
- **Insect bite-like**
- **Erythema multiforme-like**

Pruritus without visible lesions may rarely occur, but a macular variant has not been clearly documented.

The **papular form** is characterized by small or large, discrete or confluent lesions that often cluster. This is followed in frequency by the papulovesicular and plaque types; other forms are uncommon.

An **eczematous variant** has been reported, but such cases likely represent mild chronic actinic dermatitis or photo-exacerbated seborrheic or atopic eczema rather than true PMLE.

Different morphologic variants can occur simultaneously at different body sites in the same individual. For example, diffuse facial erythema and swelling may accompany typical papular lesions elsewhere.

A variant known as **benign summer light eruption**, recognized in continental Europe, refers to small papular PMLE that spares the face and develops after several days of continued sun exposure.

Laboratory Tests

Histology

Histopathologic findings in PMLE are characteristic but not pathognomonic and vary with lesion morphology. Common features include:

- Moderate to dense **perivascular lymphocytic infiltrate** in the upper and mid dermis
- Predominance of **T cells**, with occasional neutrophils and rare eosinophils
- **Upper dermal and perivascular edema**
- **Endothelial cell swelling**

Epidermal changes may be absent or include:

- Variable **spongiosis**
- Occasional **dyskeratosis**
- **Exocytosis** of lymphocytes
- **Basal cell vacuolization**

Blood Tests

Recommended laboratory evaluations include:

- **Antinuclear antibody (ANA)** and **extractable nuclear antigen (ENA) antibody** testing to exclude subacute cutaneous or other forms of lupus erythematosus
- **Red blood cell protoporphyrin** measurement if erythropoietic protoporphyria is suspected, particularly in cases with diagnostic uncertainty

Phototesting

- **Monochromator phototesting** confirms photosensitivity in up to 50% of PMLE cases but lacks specificity for distinguishing PMLE from other photodermatoses.
- **Provocation testing** using a solar simulator or broadband light source—sometimes repeated over 2–3 consecutive days—can often reproduce the eruption. This may aid diagnosis and allow biopsy when clinical and historical features are inconclusive.

Differential Diagnosis

See Box 90-1.

Complications

While most patients with PMLE do not develop other systemic diseases, a small subset may go on to develop **lupus erythematosus**. Additionally, there is an increased prevalence of prior PMLE among patients diagnosed with lupus, suggesting a possible association between the two conditions.